1. Inventory of the Zinc

When the warehouse tanks start to empty, zinc's price curve bends sharply upward. Geman & Smith spell this out: "We find a strong non-linear relationship between the adjusted-spread of the forward curve (based on a ratio between spot and futures prices) and inventory." [1]

2. Global mine production

Mine output moves slowly, so even small changes in tonnage can trigger big price swings. Fernández reports that "the percentage changes of real prices during expansions and contractions considerably exceeded those of world mine production … zinc included." [2]

3. Treatment charges

When smelters talk about rising costs, they point first to treatment charges (TCs). Tan notes that the Imperial Smelting Process "consumes a large amount of coke, which is not only expensive, but also has high CO_2 emission," making TCs the clearest signal of supply tightness. [3]

4. Global manufacturing Purchasing Managers Index

Zinc demand tracks factory activity almost one-for-one. Canuto's chart puts it simply: "Copper prices and global manufacturing PMI move together; the same pattern holds for … zinc." [4]

5. DXY

Because zinc is priced in dollars, a stronger greenback usually drags the metal lower. Yadav & Giri confirm a "strong inverse relationship between the US dollar and commodity prices," with correlation coefficients as steep as –0.72. [5]

6. Correlation between energy and Zinc prices

Energy shocks ripple straight into industrial metals. Dutta's GARCH-jump study finds "a significant price spill-over running from the oil market to the industrial metal sector," zinc included. [6]

Literature

[1] Geman H., Smith W.O. "Theory of Storage, Inventory and Volatility in the LME Base Metals." *Resources Policy* 38 (2013)

[2] Fernández V. "Assessing Cycles of Mine Production and Prices of Industrial Metals." *Resources Policy* 63 (2019)

[3] Tan P. "Challenges to Treat Complex Zinc Concentrate and Latest Technical Development." *J. Phys. Conf. Ser.* 2738 (2024).

[4] Canuto O. "Are We on the Verge of a New Commodity Super-Cycle?" (ResearchGate working paper, 2021).

[5] Yadav P., Giri J.N. "Currency Fluctuations and Commodity Prices." *ShodhKosh* 4 (2023).

[6] Dutta A. "Impacts of Oil Volatility Shocks on Metal Markets: A Research Note." *Resources Policy* 55 (2018)